MIL-B-44066A
30 September 1983
SUPERSEDING
MIL-B-44066
30 December 1982

MILITARY SPECIFICATION

BEANS WITH TOMATO SAUCE, THERMOSTABILIZED,

FOR MEAL, READY-TO-EAT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers thermostabilized beans with tomato sauce in flexible pouches for use as a component of the meal, ready-to-eat, individual.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATION

MILITARY

MIL-P-44073 - Packaging and Thermoprocessing of Foods in Flexible Pouches

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-900 - Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milk and Sugars Used in the Preparation of Canned Foods for the Armed Forces

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8940

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(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

OTHER GOVERNMENT DOCUMENTS

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal, Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

U.S. Standards for Beans

(Application for copies should be addressed to the Chief, Fresh Products Inspection, Fruit and Vegetable Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

U.S. Standards for Grades of Canned Dried Beans
U.S. Standards for Grades of Canned Tomato Juice
U.S. Standards for Grades of Concentrated Tomato Juice
U.S. Standards for Grades of Canned Tomato Puree (Tomato Pulp)
U.S. Standards for Grades of Tomato Paste

(Application for copies should be addressed to the Chief, Processed Products Branch, Fruit and Vegetable Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Interim Primary Drinking Water Regulations

(Application for copies should be addressed to the Office of Drinking Water WH550, Environmental Protection Agency, Washington, DC 20460.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN DEHYDRATED ONION AND GARLIC ASSOCIATION (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products

(Application for copies should be addressed to the American Dehydrated Onion and Garlic Association, 251 Kearney Street, No. 410, San Francisco, CA 94108.)

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(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited, herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.4, 6.1 and 6.3).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent and insect infestation, extraneous material, off-flavors, off-odors and off-colors.

3.2.1 Dry beans. The beans shall be pea beans or small white beans that conform to the requirements of U.S. No. 1 grade or higher as defined in the U.S. Standards for Beans.

3.2.2 Tomato product. The tomato product shall be juice, paste, or puree and shall be used in such quantity as to be equivalent to tomato puree of 8.0 percent natural tomato soluble solids in the amount specified.

3.2.2.1 Juice, canned. Canned tomato juice shall be U.S. Grade A of U.S. Standards for Grades of Canned Tomato Juice or U.S. Standards for Grades of Concentrated Tomato Juice, as applicable.

3.2.2.2 Puree, canned. Canned tomato puree shall be U.S. Grade A of U.S. Standards for Grades of Canned Tomato Puree (Tomato Pulp).

3.2.2.3 Paste, canned. Canned tomato paste shall be U.S. Grade A of U.S. Standards for Grades of Tomato Paste.

3.2.3 Sugar. Sugar shall be white, refined, granulated cane or beet sugar of good commercial quality and shall meet the bacterial requirements of MIL-STD-900.

3.2.4 Salt. Salt shall be noniodized, white, refined, sodium chloride with or without anti-caking agent.

3.2.5 Onion powder. The dehydrated onion powder shall comply with the quality standards, grade "Fancy" of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association (ADOGA) for Dehydrated Onions and Garlic Products.

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3.2.6 Garlic powder. The dehydrated garlic powder shall comply with the quality standards, grade "Fancy" of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association (ADOGA) for Dehydrated Onions and Garlic Products.

3.2.7 Ground spices. Spices shall be ground and of food grade quality.

3.2.8 Oil, corn or cottonseed. The oil shall be bland and free from rancid, painty, musty, soapy, fishy, metallic, beany, or other foreign or undesirable odors and flavors. The oil shall be free from settlings or foreign matter of any kind.

3.2.9 Starch, waxy maize. The waxy maize starch shall be derived from waxy maize or dent corn or a combination thereof. The starch shall have been modified so its viscosity is reduced by thermal processing and it shall meet the requirements of MIL-STD-900.

3.2.10 Water. Water used for ice making, formulation, and washing shall conform to the National Interim Primary Drinking Water Regulations.

3.3 Preparation and processing.

3.3.1 Bean preparation. The beans shall be thoroughly cleaned to remove dirt, stones, loose skins, and any other extraneous material. The beans shall also be sorted to remove beans which are spotted, discolored, or affected by pathological injury. Prior to filling, the beans shall be thoroughly washed and soaked in water to 90 percent or more of their absorption capacity in such a manner as to prevent souring. Blanching of the soaked beans will eliminate some of the bitter flavors.

3.3.2 Sauce preparation. The sauce shall be formulated as follows:

Ingredient	Percent by weight
Tomato product (8.0 percent natural tomato soluble solids)	50.000 1/
Water	31.737
Sugar, white	11.600
Oil, corn or cottonseed	4.420
Salt	2.000
Onion powder	0.110
Garlic powder	0.036
Cinnamon, ground 2/	0.025
Allspice, ground 2/	0.022
Clove, ground 2/	0.022
Mace, ground 2/	0.014
Pepper, white, ground 2/	0.014

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1/ Equals 4.5 percent total solids in the formulation.

2/ Soluble spices or essential oils of an equivalent strength may be substituted.

a. If desired to assist in the filling operation, waxy maize starch in quantities up to 2 percent by weight of the sauce may be added by first mixing with the dry ingredients before being added to the sauce.

b. The prepared sauce shall be held at a temperature not to exceed 50°F for not more than 4 hours prior to filling into pouches.

* 3.3.3 Pouch filling and processing. Each pouch shall be filled and processed such as to conform to the following requirements.

a. Filling, sealing, and thermoprocessing shall conform to the class 2 requirements of MIL-P-44073.

b. Average net weight shall be 5.0 ounces minimum.

c. No individual pouch shall contain less than 4.8 ounces of product.

d. The average washed drained weight of beans shall be 3.3 ounces minimum.

e. No individual pouch shall contain less than 3.1 ounces of washed drained beans.

f. The filled and sealed pouch shall be thermoprocessed until a sterilization value (F_o) of not less than 6 has been achieved and until the beans are tender.

* 3.4 Finished product. The finished product shall meet the requirements for U.S. Grade A of the U.S. Standards for Grades of Canned Dried Beans.

3.4.1 Palatability. The product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.5 Plant qualification. The product shall be prepared, processed, and packaged under sanitary conditions in accordance with good manufacturing practice. The product shall be processed in establishments meeting the sanitary requirements of the applicable government inspection agency.

3.6 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and acceptance service. Product acceptability shall be determined by the USDA. The USDA will determine the degree of acceptance service necessary to assure compliance with the requirements of this document. The cost of grading and acceptance services performed by the USDA involving inspection, official documentation, and related services shall be borne by the contractor.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.5.1.1 Ingredient examination. Conformance of ingredients to identity and condition requirements shall be determined by examination of pertinent labels, markings, U.S. grade certificates, invoices, certificates of analysis, or other valid documents. In addition, each ingredient shall be examined organoleptically as necessary to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation and processing requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or other valid documents shall be cause for rejection of the involved product.

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4.5.3 Filled and sealed pouch inspection. Inspection of filled and sealed pouches shall be in accordance with the quality assurance provisions of MIL-P-44073.

4.5.4 Net weight inspection. The net weight of the filled and sealed pouches shall be determined by weighing each sample unit on a suitable scale tared with a representative empty pouch. Any individual net weight of less than 4.8 ounces shall be scored as a minor defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5. Results shall be reported to the nearest 0.1 ounce. In addition, the lot shall be rejected if the sample average net weight is less than 5.0 ounces.

4.5.5 Product inspection. The filled and sealed pouches shall be held for a minimum of 10 days after completion of the sealing operation to allow for equalization of the product. The pouches shall be opened and inspected in accordance with table I. The lot size shall be expressed in pouches. The sample unit shall be the contents of one pouch. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.5.

TABLE I. Product inspection 1/ 2/

Category	Defect
Major	
151	Washed drained weight of beans in a pouch is less than 3.1 ounces. 3/ 4/

1/ Product not equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an AMS agent.)

2/ Finished product not U.S. Grade A in accordance with U.S. Standards for Grades of Canned Dried Beans shall be cause for rejection of the involved lot.

3/ To determine washed drained weight, the pouch contents shall be emptied into a flat bottom container. Approximately three times the pouch volume of 70° ± 2°F water shall be added to the container so as to cover the contents. The contents and water shall be agitated such as to remove the sauce without breaking the beans. The contents shall then be poured into a U.S. Standard No. 8 sieve in a

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manner that will distribute the product over the sieve without breaking the beans. Sieve area shall be such that the distributed product does not completely cover all the openings of the sieve. The sieve shall be tilted slightly and allowed to drain for 2 minutes before determining the washed drained weight by subtracting the sieve tare weight from the gross weight. The washed drained weight shall be reported to the nearest 0.1 ounce.

4/ If the sample average washed drained weight is less than 3.3 ounces, the lot shall be rejected.

4.5.6 Packaging inspection. Inspection of packaging shall be in accordance with the quality assurance provisions of MIL-P-44073.

5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with the applicable requirements of MIL-P-44073.

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.4, and 6.3).
c. Provisions for approved preproduction samples (see 3.4.1 and 6.3).

6.2 Award of contract. Award of contracts for the product specified in this document will be limited to plants known to maintain the required sanitation conditions of 3.5.

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Changes from previous issue. The margins of this document are marked with asterisks to indicate where changes (additions, modification, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Custodians:

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Navy - SA
Air Force - 50

Review activities:

Army - MD, TS
Navy - MC, MS
DP - SS

Preparing activity:

Army - GL

Project No. 8940-0428

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-44066A	Beans With Tomato Sauce, Thermostabilized, For Meal, Ready-To-Eat

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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